



# Choupo

The Open Source Chemical Process Simulator

## Developer Guide

Choupo-dev

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## Preface — the contract this code makes with you

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If you have written CFD glue for a CFD solver, or stared too long at a black-box solver, you already know why this project exists. Every Newton step is auditable. Every dispatch is explicit. Every dependency is in the C++ standard library. The price of that discipline is that contributors have to play by a few rules. They are short, and they are non-negotiable.

This guide is opinionated on purpose. Read §1 carefully — those three principles override everything else — and treat §25 as the project’s only blunt list of don’ts.

**Independence and provenance.** Choupo is an independent project. Its file-based case organization (`controlDict`-style dictionaries, `constant/` and time directories such as `0/`) was inspired in part by the conventions popularised by OpenFOAM. Choupo is not affiliated with, endorsed by, or sponsored by OpenCFD Ltd or the OpenFOAM Foundation. The modules closest in spirit are independent implementations, not adaptations: the dictionary parser is Choupo’s own component-aware tokenizer; model selection uses explicit factory registrations (deliberately *not* runtime-selection macros, §6); the `Rosenbrock23` integrator implements the published Shampine & Reichelt (SIAM J. Sci. Comput. 18, 1997) formulas; and the time-directory writer implements the layout convention from scratch. No OpenFOAM source code was adapted or reused.

# 1 Philosophy — the contract we promise

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*What you'll learn.* The three invariants that every contribution is judged against, in plain language.

1. **Pedagogical clarity beats elegance.** The simulator must be readable by a chemical-engineering student in their third year. No template metaprogramming, no auto-magic, no clever short-circuits. Every Newton iteration and every  $K$ -value must be inspectable in both the source and the run-time output.
2. **Source transparency — no hidden registration.** Every factory base class has an *explicit* `registerBuiltins()` called from `main.cpp`. No `REGISTER_TYPE(MyClass)` macros, no static initialisers, no linker-dependent constructors. The student `grep`-searches `registerBuiltins` and *sees* every built-in type.
3. **Dependency purity — no copyleft or restrictive libraries linked into the binary.** C++ 17, the C++ standard library, and GNU make. Nothing else. No Boost, no Eigen, no Sundials, no CMake. Hand-rolled Newton, Gauss elimination, RK4, Wegstein, Michelsen.

**Key idea.** An “industrial-grade” contribution that violates any of these will be rejected even if it converges better and runs faster.

## 2 Repository tour

*What you'll learn.* What lives where. Useful as a map; you'll come back to it.

```

Choupo/
|-- Makefile                top-level: release/debug switch
|-- make/
|   |-- compiler.mk        CXX, CXXFLAGS, PLATFORM detection
|   |-- rules.mk           pattern rules + dependency tracking
|   |-- wasm.mk            Emscripten WASM build (gui/)
|-- build/<PLATFORM>[Debug]/ all.o/.d/binary (gitignored)
|-- choupoSolve             symlink -> last built binary
|-- etc/bashrc             sourceable shell env
|-- bin/                   runCase, runApplication, listCases, cleanCase
|-- data/
|   '-- standards/         the FROZEN, committee-managed reference tree
|       |-- components/    <name>.dat --- component catalogue (arity-1)
|       |-- species/       <name>.dat --- model species (ions, aq. gases)
|       |-- chemistry/     real equilibria: salts, speciation, minerals
|       |-- assets/        physical kit, FLAT with kind = consumer
|       |                  (membranes, materials, adsorbents, resins)
|       |-- utilities/     <name>.dat --- plant-utility catalogue
|       |-- parameters/    every pair catalogue (arity-2):
|       |                  NRTL/ UNIQUAC/ Wilson/ Henry/ SRK/
|       |                  Pitzer/ eNRTL/ solution/ UNIFAC/
|       '-- mixtures/      predefined mixtures (air, ...)
|-- docs/                  THIS FILE + userGuide.{md,tex,pdf}
|-- src/
|   |-- core/              Dictionary, SimulationResult, Constants, Types
|   |-- streams/           ProcessStream
|   |-- thermo/            Component, ThermoPackage, Database +
|   |                      phase/, vaporPressure/, activityCoefficient/,
|   |                      equationOfState/, heatCapacity/, membrane/
|   |-- materials/         Material, MaterialRegistry
|   |-- solver/            NewtonRaphson, NewtonND, Wegstein, StabilityTest
|   |-- unitOperations/    UnitOperation base + per-category folders
|   |-- outerDriver/       OuterDriver base + SweepDriver + FitBinaryPair
|   |-- postProcessing/    PostProcessor base + Sizing/Costing passes
|   |-- propertyOps/       PropertyOperation base (choupoProps)
|   |-- control/           Controller base + PID/Schedule (choupoCtrl)
|   |-- reporting/         BalanceMath, EnergyBalanceReport (the ledger)
|   |-- io/                SolutionWriter --- OpenFOAM-style instants
|   '-- applications/      choupoSolve / choupoBatch / choupoCtrl /
|                           choupoProps -- one main.cpp per binary
|-- tests/                 regression scripts (bin/runTests)
|-- tutorials/             end-to-end cases (~190, under
|                           steady/ batch/ ctrl/ props/ plant/)
|-- gui/                   web GUI (see CLAUDE.md Sec. 12)
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'-- README.md

```

**Key idea.** Data lives under `data/standards/`, write-guarded. The engine *refuses* to write under `data/standards/` — adding a number there is a deliberate curation act, not a side effect of a run. WHERE a number belongs (an intrinsic field in `components/<name>.dat`, a pair file in a catalogue, or an operation's `constant/`) is decided by the **data-governance doctrine** (`docs/ai/data-doctrine.md`): the canonical reference for every curation question, summarised in §21.

**Key idea.** There is *one binary per problem class*, four in all — they share the same library under `src/{core,thermo,solver,...}` and differ only in their `applications/<name>/main.cpp`:

- `choupoSolve` — steady-state simulation,  $F(\mathbf{x}) = 0$  (root-finding);
- `choupoBatch` — batch / recipe-driven, per-vessel ODE integration + time-triggered events;
- `choupoCtrl` — dynamic continuous + control,  $d\mathbf{Y}/dt = f(\mathbf{Y}, \mathbf{u}, t)$  with PID/schedule controllers;
- `choupoProps` — property evaluation + parameter fitting (a distinct problem class: neither root-finding nor integration).

We split by *problem class*, never by numerical *strategy* (no `*Foam`-style proliferation): within each binary the strategies (Newton, Wegstein, RK4, Nelder–Mead, ...) are run-time selections via dicts, not compile-time ones.

## 3 Build system

```
make all           # release, -O2, build/linux64Gcc/
make MODE=debug all # -O0 -g3, build/linux64GccDebug/
make print         # show resolved variables
make clean         # rm objs + binary for current MODE
make distclean     # rm entire build/
```

Sources are discovered with `find src -name '*.cpp'` — drop a new `.cpp` anywhere under `src/` and it picks up automatically. Header dependencies are auto-tracked via `-MMD -MP` (`.d` files alongside each `.o`).

Platform string is derived from `uname -s -m`. Currently only `linux64Gcc` is exercised; `darwin64Clang` and `linux64Clang` should “just work” with the same Makefile (modulo `<filesystem>` linking on older toolchains — see §17).

The WASM build (`make wasm`) is a sibling target — see `make/wasm.mk` and `CLAUDE.md` §13 for the Emscripten quirks.

## 4 Build architecture: shared library, optional modules, case code

Since the build is *dynamic* and *modular*. Three forces drove it: incremental compilation (change one binary, don't rebuild the engine), a clean extension story (users add unit ops without touching `src/`), and — load-bearing for the modular architecture — making it hard to produce a standalone binary that carries the whole engine inside. The model: an engine built as a shared library, and small binaries that *reference* it.

**Key idea.** A binary never embeds the engine; it links `libchoupo.so` at build time and finds it at run time via `rpath`. A binary copied to a machine without the library simply does not start. This is the same model — one engine, many small binaries that share it, so rebuilding a binary never rebuilds the engine.

### 4.1 The shared engine library (`libchoupo.so`)

`make lib` archives every engine object (everything under `src/` except the per-application entry points) into a single shared object:

```
make lib                # -> build/<plat>/libchoupo.so    ($(CXX) -shared)
```

The engine is compiled `-fPIC` (position-independent) on Linux. Each binary then links `-lchoupo` with `rpath` set to `$ORIGIN` — the binary's own directory — which is where the `.so` is built; the project-root symlinks resolve to the real binary location, so `$ORIGIN` still points at the `.so`. The result is tiny binaries:

| Artefact                           | Size    | Contains the engine?    |
|------------------------------------|---------|-------------------------|
| <code>libchoupo.so</code>          | 1.7 MB  | it <i>is</i> the engine |
| <code>choupoSolve</code> (binary)  | 86 KB   | no — references it      |
| <code>choupoCase</code> (a user's) | ~100 KB | no — references it      |

**Common pitfall.** The `win64MinGW` cross-build keeps *static* linking on purpose: the generated `.exe` is self-contained. The `LIB_DEP` / `LIB_LINK` Makefile variables carry the platform branch (objects directly for `win64MinGW`, `-lchoupo` on Linux).

### 4.2 Optional modules — a clean modular split

The engine can carry OPTIONAL modules as separate `.so` files, so a slim core (the teaching engine) builds without them. The membrane module is Vitor's NF/RO research code — conceptually distinct from the core and kept in its own library so a core-only build is possible. Everything is GPL-3.0-or-later; the split is modular, not a licence boundary:

| Library                           | Contents                                          | Licence          |
|-----------------------------------|---------------------------------------------------|------------------|
| <code>libchoupo.so</code>         | core (thermo, solver, unit ops, reports)          | GPL-3.0-or-later |
| <code>libchoupoMembrane.so</code> | NF/RO membrane ( <code>spiralWoundModule</code> ) | GPL-3.0-or-later |

The mechanism is the same link-time stub-swap used for user code (§4.3), applied to a module:

1. The optional unit op lives under its own folder (`src/unitOperations/membrane/`); its objects are filtered *out* of `LIB_OBJS` and compiled into `libchoupoMembrane.so` instead.
2. It is **not** registered in `UnitOperation::registerBuiltins()` (the core must not depend on it). Instead a `registerMembrane.cpp` in the module defines a `registerModules()` hook that registers the type via the usual explicit factory.
3. `choupoSolve/main.cpp` calls `registerModules()` right after `registerBuiltins()`. The core binary links an empty `registerModulesStub.cpp` *only* when the module is absent; the Makefile (`MEMBRANE_PRESENT := $(wildcard src/unitOperations/membrane)`) links exactly the `.so` *or* the stub, never both.

```
# slim core-only build: remove the module source, rebuild
rm -rf src/unitOperations/membrane && make all
# -> only libchoupo.so is produced; the membrane type is unknown:
#    "unknown type 'spiralWoundModule'"
```



To add another optional module, do exactly what the membrane does: a folder under `src/`, kept out of `registerBuiltins()`, with its own `registerModules()` compiled into its own `.so`. (When a *second* module appears, generalise the single `registerModules()` hook into a small list — still explicit, still no auto-registration.)

### 4.3 Case-local code (case/code/)

A user — a student, a customer — may write their own C++ under a case's `code/` folder and have it compiled into a per-case binary. Build-time only: **no runtime dlopen**, **no macro self-registration**, faithful to the glass-box credo.

```
case/
  code/
    MyOp.H MyOp.cpp      # the user's unit op (derives UnitOperation)
    registerUserTypes.cpp # defines registerUserTypes() -> registerType("myOp",...)
    options              # OPTIONAL: extra -I / -l / flags (see below)
  system/ constant/
```

**What code/ can register.** The primary, tutorialised use is a new *unit operation*. But `registerUserTypes()` runs *after* every `registerBuiltins()`, so it can register **any type with an explicit factory** — the same hook, the same pattern:

| What                                                      | via                                             | Status                        |
|-----------------------------------------------------------|-------------------------------------------------|-------------------------------|
| unit operation                                            | <code>UnitOperation::registerType</code>        | demonstrated<br>(userOp01/02) |
| thermo model ( $\gamma$ , EoS, $P^{\text{sat}}$ , $c_p$ ) | <code>ActivityModel::registerModel</code> , ... | possible, no tutorial yet     |
| outer driver (sweep / optim / DesignSpec)                 | <code>OuterDriver::registerType</code>          | possible                      |
| report / sizing / costing                                 | <code>Report::registerType</code> , ...         | possible                      |

**What the user writes.** Three files in `code/`: the unit op (`MyOp.{H,cpp}`, a normal class deriving `UnitOperation` with a `solve()` override), and `registerUserTypes.cpp`, which puts the class in the factory:

```
// case/code/registerUserTypes.cpp
void registerUserTypes() {
    Choupo::UnitOperation::registerType(
        "myOp", [] { return std::make_unique<Choupo::MyOp>(); });
}
```

**How it is compiled.** `bin/buildCode` runs one ordinary `g++` (it prints the exact command — nothing hidden):

```
g++ -std=c++17 -O2 -I src \           # (1) the engine's headers
    build/<plat>/.../choupoSolve/main.o \ # (2) the engine main() (calls
                                         # registerUserTypes)
    case/code/registerUserTypes.cpp \   # (3) the user's registration
    case/code/MyOp.cpp \                # (4) the user's unit op
    -Lbuild/<plat> -lchoupo [-lchoupoMembrane] \ # (5) engine.so (+ optional module)
    -Wl,-rpath,build/<plat> \           # (6) where to find the.so at run time
    -o case/code/.build/choupoCase     # (7) the per-case binary (~100 KB)
```

The user's `.cpp` are compiled fresh each time (their `.o` are temporaries in `/tmp`, dropped after linking — only the binary remains). The binary links the engine `.so` DYNAMICALLY (5): it references the engine, never embeds it.

**The trick — who supplies `registerUserTypes()`.** The engine's `main.o` calls `registerUserTypes()` (right after `registerBuiltins()`) but does *not* define it — an unresolved symbol. The linker resolves it from exactly one place, and that place is the build-time choice:

| Binary                             | <code>registerUserTypes()</code> from          | Result                      |
|------------------------------------|------------------------------------------------|-----------------------------|
| stock <code>choupoSolve</code>     | <code>registerUserTypesStub.cpp</code> (empty) | registers nothing           |
| the case's <code>choupoCase</code> | <code>case/code/registerUserTypes.cpp</code>   | registers <code>myOp</code> |

In the `buildCode` command above the stub is *not* linked — only the user's file is — so the linker resolves the symbol with the user's definition. Plain symbol resolution, no weak symbols, no `dlopen`, no macros. (Optional modules use the same pattern with `registerModules()`, §4.2.)

### How it is used — at run time.

```
choupoCase runs:
main()
-> UnitOperation::registerBuiltins() // cstr, flash, mixer,...
-> registerModules()                // optional modules (membrane.so)
-> registerUserTypes()             // THE USER'S: registry()["myOp"] = constructor
Flowsheet reads type myOp; in flowsheetDict
-> UnitOperation::New("myOp")       // look up in the std::map
-> the user's constructor -> object -> the user's solve()
```

The factory is just a `std::map<string, function>`; `registerType` adds an entry, `New` looks one up. `myOp` sits beside the built-in types — no special case.

**What `solve()` receives — streams and thermo.** The op's whole world arrives through two arguments:

```
int solve(const DictPtr& dict, const ThermoPackage& thermo, int verbosity);
```

*Streams (the dict).* Choupo is sequential-modular: an op does NOT rummage a global stream registry. The *Flowsheet* *injects* the unit's inlet stream into the `dict` as sub-dicts, plus the unit's own *operation* block:

```
auto feed = dict->subDict("feed");           // F (kmol/s SI), T, P of the inlet
auto comp = dict->subDict("composition");     // mole fractions z_i
auto oper = dict->subDict("operation");       // YOUR parameters (from flowsheetDict)
```

You read your inputs, compute, and publish outputs into `produced_`; the *Flowsheet* binds them to the next units by the *outputs* ( ... ) order. (For several inlets, the unit declares *inputs* (a b ) and reads each.)

*Thermodynamics (the thermo).* The `ThermoPackage&` gives full access to whatever models the case configured — you do not need to know *which* ( $\gamma$  = NRTL? EoS = SRK?); you call the method and the selected model answers:

```
thermo.n(); thermo.indexOf("benzene"); thermo.comp(i); // count, index, Component
thermo.comp(i).vp().Psat_Pa(T);                       // vapour pressure (Antoine/...)
thermo.Kvec(T, P, x, y);                               // K-values (the case's gamma-phi)
thermo.activity().gamma(T, x);                         // activity coefficients gamma_i
thermo.Hliquid(T, x); thermo.Hvapour(T, y);            // sensible-datum enthalpies
thermo.H_real(T, P, y);                               // real gas (H_ig + EoS departure)
thermo.H_stream_formation(T, P, vf, z);               // elements-datum (carries Hf)
```

So a user op can flash, balance energy, or solve an equilibrium with the case's thermo — exactly what the built-in CSTR / flash / column do; it is a first-class citizen, not an add-on.

**Common pitfall.** Transport properties (viscosity, conductivity, diffusivity) do not exist yet — they are on the roadmap as a `TransportModel` with a factory. When they land they will be reachable through the same `thermo` handle; today a user op has thermodynamics only.

| Command                             | What it does                                                                                                             |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <code>buildCode &lt;case&gt;</code> | <i>compiles only</i> (shows the exact g++ command, reports compiler errors); the development-loop command                |
| <code>runCase &lt;case&gt;</code>   | <i>runs</i> the case — and if it has a <code>code/</code> folder, compiles it first (visibly) via <code>buildCode</code> |

Both step up automatically if invoked from inside the `code/` folder. External dependencies (a property library, GSL, Eigen, ...) go in an optional `code/options` file:

```
# case/code/options
INCLUDE = -I/opt/coolprop/include
LIBS    = -L/opt/coolprop/lib -lCoolProp
FLAGS   = -O3
```

**Common pitfall.** User code lives in `case/code/`, **never** in `src/`. Three reasons: isolation (the engine tree stays auditable), authorship / licence (`src/` is GPL-3.0-or-later; the user owns their `code/`), and pedagogy (a clear engine-vs-mine boundary). Those external libraries are the *user's* dependencies on the *user's* code — the engine itself stays dependency-free.

## 5 Core types

*What you'll learn.* Four types live in `src/core/` and `src/streams/` and propagate through every layer of the simulator: get familiar with them before touching anything else.

### 5.1 Dictionary

hierarchical plain text. Every case dict is parsed into a `Dictionary`. Key operations:

```
auto d = Dictionary::fromFile("system/flowsheetDict");

scalar T = d->lookupScalar("T");           // raw
scalar P = d->lookupScalar("P", Dims::pressure); // dim-checked
auto sub = d->subDict("operation");
auto units = d->lookupDictList("units");
auto comps = d->lookupWordList("components");

// Path-based mutation --- used by outer drivers
d->setScalarAtPath("units[0].operation.refluxRatio", 3.2);
```

The tokenizer treats [ and ] as word chars (so paths work). Optional `FoamFile` headers are skipped silently for compatibility with files from other tools.

**Three input forms for scalars.** The parser accepts named-unit (`P 1 bar;`), the bracket (`A_w [-1 2 1 0 0] 1.5e-12;`), and raw SI (`R 0.5;`). Named and bracket forms tag the entry with its physical dimensions [M L T Theta N] so the dim-checked `lookupScalar(key, expectedDims)` overload can throw a clear error on mismatch. Raw entries pass unchecked — the caller is asserting the value is canonical SI.

Available named dimensions live under `Dims::` in `src/core/Dimensions.H` (`pressure`, `molarFlow`, `massFlow`, `temperature`, `length`, `area`, `volume`, `concentration`, `molarHeatCap`, `viscosity`, `diffusivity`, `permeabilityWater`, `massTransferCoeff`,...). Adopt them at the call sites where the unit op reads its operating parameters.

### 5.2 ProcessStream

What travels between unit operations (units: SI canonical):

```
struct ProcessStream {
    std::string      name;
    scalar           F;      // total FLUID molar flow [kmol/s]
    scalar           T;      // K
    scalar           P;      // Pa
    std::vector<scalar> z;    // FLUID mole fractions, size = NC
    scalar           vf;     // vapour fraction (VLE outputs)
    std::vector<scalar> s;    // solid molar flow per component [kmol/s]
    ParticleSizeDist psd;     // size distribution of the solid phase
    std::string      category; // "" = process; non-empty = utility leg
};
```

**Key idea.** `F`, `z` and `vf` describe the *fluid* phases only; the **solid is carried alongside in `s[]`** (kmol/s per component, empty or all-zero = no solids), with its size distribution in `psd`. A crystalliser writes its crystals into `s[]`; the balance machinery (`src/reporting/BalanceMath.H`) reads `s[]` so mass *and* energy close over the solid too (see §22). The `category` tag (populated by utility `<name>;`) lets the energy ledger separate a utility leg — heat crossing the boundary — from a process stream.

Mass-basis flow is a derived view, not a stored field — if you need kg/s, call `F_mass(stream, thermo)` from `src/streams/StreamMass.H`, which computes  $F \cdot \sum_i z_i MW_i$  on demand. One source of truth for the flow (molar); mass cannot drift out of sync.

When you produce streams in your unit-op, fill *all* fields. At print sites you typically scale by 3600 for kmol/h and by  $10^{-5}$  for bar, or hand the value plus its dimensions to `DisplayUnits::instance().format(v, dims)` which honours the case's `controlDict.units` preferences.

### 5.3 SimulationResult

What a single pass produces — input to post-processors and outer drivers:

```
struct SimulationResult {
    std::map<std::string, ProcessStream>    streams;
    std::map<std::string, std::map<std::string, scalar>> kpis;
    std::map<std::string, EquipmentSizing>  sizings;      // SizingPass
    std::map<std::string, CostBreakdown>    costs;        // CostingPass
    bool                                    converged;
};
```

The `kpis` map is keyed `unitName → { kpiName → value }`. Outer drivers read it; post-processors append to `sizings` / `costs`. (The `sizings` field was named `dimensions` until; renamed because `Dimensions` is now the physical-dimensions tag.)

### 5.4 ThermoPackage

Holds:

- components (vector of `Component`),
- one or more `Phase` instances (vapor / liquid / solid),
- convenience getters: `nComponents()`, `component(i)`, `phase(name)`, `Psat(i, T)`, `gamma(phase, x, T)`, etc.

Every unit operation receives a `const ThermoPackage&` in `solve()`.

**Two enthalpy datums — and why only one survives.** The package exposes enthalpy on two references:

- `Hliquid(T,x)` / `Hvapour(T,y)` — the **sensible** datum, anchored at an arbitrary  $T_{\text{ref}}$ . Fine for a single-stream heat-up, useless across a reactor: it silently drops the heat of reaction.
- `H_stream_formation(T,P,vf,z)` — the **elements** datum (formation enthalpies at 298.15 K, the same reference as the ideal-gas  $H_{ig}$ ). The reaction enthalpy then *emerges* as the difference of formation enthalpies, so a balance closes *through* a reactor, a crystalliser, or a heat of solution.

**Key idea. One datum, no fallback (§22).** The energy-balance ledger runs on the elements datum *only*. The former silent “fall back to the sensible datum” arm was **deleted**: a component missing its `standardThermochemistry` now *throws*, naming the component, so a curation gap is loud, never a quietly wrong closure.

## 6 The factory pattern — explicit, no macros

*What you'll learn.* The single registration mechanism used everywhere in the code, and why we refuse the macro-based alternative.

Every extensible base class follows this template:

```
class UnitOperation {
public:
    using Factory = std::function<std::unique_ptr<UnitOperation>()>;

    static void registerType(const std::string& name, Factory f);
    static std::unique_ptr<UnitOperation> New(const std::string& type);
    static std::vector<std::string> availableTypes();
    static void registerBuiltins();           // <-- called in main()

    // ---- virtual interface ----
    virtual const std::string& type() const = 0;
    virtual int solve(const DictPtr&, const ThermoPackage&, int verbosity) = 0;
    virtual std::vector<ProcessStream> producedStreams() const { return {}; }
    virtual std::map<std::string, scalar> kpis() const { return {}; }

protected:
    // helpers...
};
```

`registerBuiltins()` (in the corresponding `.cpp`) is the single place that *lists every type*:

```
void UnitOperation::registerBuiltins()
{
    auto reg = [](const std::string& n, Factory f) { registerType(n, f); };

    reg("isothermalFlash", []{ return std::make_unique<IsothermalFlash>(); });
    reg("flash",          []{ return std::make_unique<IsothermalFlash>(); });
    reg("adiabaticFlash", []{ return std::make_unique<AdiabaticFlash>(); });
    reg("bubbleT",        []{ return std::make_unique<BubblePoint>(); });
    reg("dewT",           []{ return std::make_unique<DewPoint>(); });
    reg("cstr",           []{ return std::make_unique<CSTR>(); });
    reg("pfr",            []{ return std::make_unique<PFR>(); });
    reg("heatExchanger",  []{ return std::make_unique<HeatExchanger>(); });
    reg("HX",             []{ return std::make_unique<HeatExchanger>(); });
    reg("mixer",          []{ return std::make_unique<Mixer>(); });
    reg("splitter",       []{ return std::make_unique<Splitter>(); });
    reg("distillationColumn", []{ return std::make_unique<DistillationColumn>(); });
    reg("column",         []{ return std::make_unique<DistillationColumn>(); });
}
```

`main.cpp` calls each `registerBuiltins()` once at startup, in order. That is the *entire* registration story. No magic, no surprises.

**Common pitfall.** Do *not* introduce `REGISTER_TYPE(...)` macros, anonymous-namespace static instances, or any pattern that runs registration as a side effect of TU loading. We have rejected this approach explicitly because the linker can drop unused TUs and the failure mode is silent.

## 7 Adding a new unit operation

*What you'll learn.* The five moving parts in outline — skeleton, implementation, registration, tutorial, regression. The full *tim-tim-por-tim-tim* walkthrough, with a real worked example and the mandatory WASM rebuild, is in §23.1; this is the at-a-glance version.

### 7.1 Step 1 — Skeleton

Create `src/unitOperations/<category>/MyOp.H`:

```
#pragma once
#include "core/Dictionary.H"
#include "thermo/ThermoPackage.H"
#include "unitOperations/UnitOperation.H"

namespace Choupo {

class MyOp : public UnitOperation {
public:
    MyOp() = default;
    const std::string& type() const override { return type_; }

    int solve(const DictPtr& dict,
              const ThermoPackage& thermo,
              int verbosity) override;

    std::vector<ProcessStream> producedStreams() const override { return produced_; }
    std::map<std::string, scalar> kpis() const override { return kpis_; }

private:
    inline static const std::string type_ = "myOp";
    std::vector<ProcessStream> produced_;
    std::map<std::string, scalar> kpis_;
};

} // namespace Choupo
```

### 7.2 Step 2 — Implementation

`src/unitOperations/<category>/MyOp.cpp` — implement `solve()`. Boilerplate to follow:

```
int MyOp::solve(const DictPtr& dict, const ThermoPackage& thermo, int verbosity)
{
    // 1. The Flowsheet INJECTS the inlet + your params as sub-dicts of 'dict'
    //     (sequential-modular: no global stream registry to rummage).
    auto feed = dict->subDict("feed");           // F (kmol/s SI), Tfeed, Pfeed
    auto comp = dict->subDict("composition");     // mole fractions z_i
    auto op = dict->subDict("operation");         // YOUR parameters
    const scalar F = feed->lookupScalar("F", Dims::molarFlow);
    const scalar Tfeed = feed->lookupScalar("Tfeed", Dims::temperature);
    const scalar T = op->lookupScalar("T", Dims::temperature);

    // 2. Echo inputs (verbosity >= 3 --- the pedagogical default)
    if (verbosity >= 3)
        std::cout << " [MyOp] T = " << T << " K, F_in = "
                    << (F * 3600.0) << " kmol/h\n"; // print in kmol/h, store SI

    // 3. Solve the model --- use solver/ classes for Newton/Wegstein.
    //     newton1D(f, dfdx, x0, NROptions) -> NRResult { root, iterations, converged }
    solver::NROptions nro; nro.lower = 200.0; nro.upper = 700.0;
    auto r = solver::newton1D([&](scalar x){ /* g(x) */ return 0.0; },
                             [&](scalar x){ /* g'(x) */ return 1.0; },
```

```

        /*x0=*/Tfeed, nro);

// 4. Build produced streams into produced_ (order = your contract;
//    the Flowsheet binds them to names via 'outputs (...)')
produced_.clear();
ProcessStream out;
out.name = "myOp_out";
out.F = F; out.T = T; out.P = feed->lookupScalar("Pfeed", Dims::pressure);
out.z.resize(thermo.n()); // fill ALL fields
produced_.push_back(out);

// 5. Populate KPIs (they feed sizing / costing / sensitivity)
kpis_.clear();
kpis_["T_out"] = T;
kpis_["F"] = F;

return 0; // 0 == converged
}

```

### 7.3 Step 3 — Register

Edit `src/unitOperations/UnitOperation.cpp`:

```

#include "<category>/MyOp.H"
//...
void UnitOperation::registerBuiltin() {
    //...
    reg("myOp", []{ return std::make_unique<MyOp>(); });
}

```

### 7.4 Step 4 — Tutorial

Create `tutorials/myOp01_quick_smoke/` with `system/controlDict`, `system/flowsheetDict`, and `constant/thermoPhysPropDict`. Keep the case trivial so it doubles as a regression test.

### 7.5 Step 5 — Build, WASM, regression

```

make all # recursive find picks the new .cpp up automatically
make wasm-gui # MANDATORY: the GUI's WASM is a SEPARATE build (see below)
bin/runTests # NaN/inf guard on every case + golden-master KPIs

```

**Common pitfall.** Rebuild the WASM or the browser will not see your type. The WASM solver is a separate compile from the native binary. Symptom of a stale `.wasm`: the native run works, but the GUI errors `UnitOperation::New: unknown type 'myOp'`. Fix with `make wasm-gui` — the incremental build tracks sources, flags and the standards tree by content, so a changed input always rebuilds; if it ever says “Nothing to be done” after a real change, that is a dependency-graph bug to report, and `make wasm-clean` is a deliberate diagnostic, not a ritual. Never validate with a bare `choupoSolve > /dev/null && PASS` loop — a solver can return NaN with exit code 0; `bin/runTests` is the only honest gate (§16).

All previously-passing tutorials *must* still pass.



## 8 Declaring what a unit does to material

Some engine machinery needs to know not what a unit *is* but what it *does to the material passing through it*. The carried species inventory is the current example: when a stream arrives with a `speciation {}` block and the unit resolves no chemistry of its own, the block has to cross somehow, and how depends entirely on the unit's material mapping.

A splitter is the easy case and the instructive one. It divides matter by *one* fraction common to every species, so a carried species inventory — which is extensive — divides by exactly that fraction. Algebra, not chemistry. A separator, membrane, filter or adsorber has no such fraction: its outlets are not proportional copies of its inlet, and the inventory cannot be divided without a model.

The distinction is **declared**, on the unit:

```
// UnitOperation.H -- the default is silence
virtual std::string materialMapping() const { return ""; }

// Splitter.H -- the capability, stated
std::string materialMapping() const override
{ return "proportionalExtensiveSplit"; }
```

**Key idea.** The name of the class is irrelevant; what is declared is what the contract reads. It would have been shorter to test whether the unit is a `Splitter` — and it would have been wrong the day somebody wrote a selective splitter, or a proportional divider that is not called one. A capability that a caller must rely on is declared by the class that has it, never inferred by the caller from a name.

Two consequences worth stating, because the first instinct on both was the wrong one:

- **Refusing was considered and rejected.** The engine already knows the fraction; refusing would force an author to restructure a flowsheet around a limitation that does not exist.
- **So was falling back to re-derivation.** Letting the post-solve pass rebuild the two branch blocks from scratch would replace known arithmetic with a model-dependent answer — worse than either alternative.

What must close, and is checked: the two branch inventories sum back to the inlet's, species by species, and each branch stays electroneutral, because one common fraction cannot break a balance that held before it. `basis01_two_unit_chain` exercises exactly this on the far side of a model boundary.

When you add a unit whose outlets *are* proportional copies of its inlet, declare the mapping. When they are not, say nothing — the default silence is the honest answer, and the machinery that needs the capability will decline rather than guess.

## 9 Adding a new thermodynamic model

Same explicit-factory spirit as everywhere else, with `ActivityModel` / `EquationOfState` / `VaporPressureModel` / `HeatCapacityModel` / `Phase` as the base. The one difference from `UnitOperation` is **the factory takes construction arguments** (a thermo model needs the configuring sub-dict, and usually the component list), so its `Factory` signature is not the nullary unit-op one — match the base you are extending:

| Base                            | Factory / register call                                                                                  |
|---------------------------------|----------------------------------------------------------------------------------------------------------|
| <code>VaporPressureModel</code> | <code>(const DictPtr&amp;); registerModel(name, f)</code>                                                |
| <code>HeatCapacityModel</code>  | <code>(const DictPtr&amp;); registerModel(name, f)</code>                                                |
| <code>ActivityModel</code>      | <code>(const DictPtr&amp;, const std::vector&lt;std::string&gt;&amp;);<br/>registerModel(name, f)</code> |
| <code>EquationOfState</code>    | <code>(const DictPtr&amp;, const std::vector&lt;Component&gt;&amp;); registerModel(name,<br/>f)</code>   |

The detailed, copy-pasteable recipe — with the validation-first AAD step and the EoS data-doctrine path — is in §23.2. The short version, adding UNIQUAC as an activity model:

1. `src/thermo/activityCoefficient/UNIQUAC.{H,cpp}` inheriting `ActivityModel`. Override the pure virtuals `gamma(scalar T_K, const sVector& x), n(), modelName()`; a constructor `UNIQUAC(const DictPtr&, const std::vector<std::string>&)` reads the pair parameters.
2. In `src/thermo/activityCoefficient/ActivityModel.cpp`, add the `#include` and *one* line in `registerBultins()`:

```
#include "UNIQUAC.H"
//...
void ActivityModel::registerBultins() {
    registerModel("ideal", [](const DictPtr&, const std::vector<std::string>& n)
        -> std::unique_ptr<ActivityModel>
        { return std::make_unique<IdealSolution>(n.size()); });
    registerModel("NRTL", [](const DictPtr& d, const std::vector<std::string>& n)
        -> std::unique_ptr<ActivityModel>
        { return std::make_unique<NRTL>(d, n); });
    registerModel("Wilson", [](const DictPtr& d, const std::vector<std::string>& n)
        -> std::unique_ptr<ActivityModel>
        { return std::make_unique<Wilson>(d, n); });
    registerModel("UNIQUAC", [](const DictPtr& d, const std::vector<std::string>& n) // new
        -> std::unique_ptr<ActivityModel>
        { return std::make_unique<UNIQUAC>(d, n); });
}
```

3. Use it in a tutorial's `thermoPhysPropDict` (inside `equilibrium.liquid`):

```
activityModel { model UNIQUAC; binaryParameters {... } }
```

## 10 Property operations (choupoProps)

`choupoProps` runs *property operations* rather than a flowsheet. Each inherits `PropertyOperation` (same explicit factory as everywhere else; `registerBuiltins()` is called from `src/propertyOps/PropertyOperation.cpp`). An op reads a `state` ( $T$ ,  $P$ , composition), a `properties` list and an `output` block, and writes a CSV the GUI auto-plots.

*Which ops exist is not listed here.* This paragraph used to carry seven names and the engine registers twenty-seven; a list of built-ins in a manual is a second home for a fact the code already states, and it drifts the first time somebody adds one. The authority is `gui/schemas/operations/`, and the props guide's operations chapter is *rendered* from it (`bin/curate/props_ops_reference.py`, with a `runTests` gate that refuses a stale render). The props-unit partition is derived too: an op is a unit op precisely when one of the three `UnitOperation` registries registers it.

Property keywords live in `PropertyEvaluator.cpp`: per-component `Psat_<c>`, `gamma_<c>`, `y_eq_<c>`, `Cp_liquid_<c>` (each guarded — e.g. `Psat` on a component with no vapour-pressure model throws a clear error instead of dereferencing null), and mixture scalars `Z`, `v_molar`, `H_real`, `S_real`, transport, etc. `PropertyScan1D` wraps each evaluation in `try/catch`, writing `nan` on failure so one undefined component never aborts the whole scan.

**propertyScanTernary — ternary phase diagrams by reuse.** Walks the composition simplex ( $x_1 + x_2 + x_3 = 1$ ) at fixed  $(T, P)$  and, at every node, calls the SAME solver the `isothermalFlash` unit op calls — so the diagram cannot disagree with a single flash. Three modes:

- `bubbleT` — a boiling-temperature *surface*: per node `BubblePoint::compute`; needs only the three vapour pressures (ideal binaries are fine), no liquid-liquid data.
- `lle` (default) — a *solubility* map: the liquid-liquid flash (`PhaseSet::LL`) gives a clean binodal + tie-lines, no spurious subcooled vapour (the extraction/decanter view).
- `vll` — the full vapour + two-liquid classification (`PhaseSet::VLLE`).

Each simplex node is independent, so the scan is embarrassingly parallel: the optional `shard { k; n; }` block computes only rows with  $i \bmod n = k$  (round-robin, balanced). The GUI fans  $N = \text{cores} - 2$  shards across  $N$  WASM workers and merges the partial CSVs (the union of shards reproduces the full grid exactly). No physics is reimplemented — the op is a triangular loop + a classifier + a CSV writer over `solveCore/BubblePoint`.

**Adding a property op.** Mirror the recipe of §6: implement `src/propertyOps/MyOp.{H,cpp}` inheriting `PropertyOperation` (override `type()`, `run(dict,thermo,verbosity)`, optionally `diagnostics()`), then one `reg("myOp", ...)` line in `PropertyOperation::registerBuiltins()`. Rebuild the native binary AND the WASM (`make wasm-gui`) or the browser will not see the new op.

**And write the schema** — `gui/schemas/operations/myOp.schema.json`, with a `title` and a `description` for the op and for every key it reads. It is not optional decoration: the GUI builds its form from it, `check_schema_coverage` refuses a key the corpus uses and the schema does not declare, and the props guide's reference chapter is generated from it. An op without a schema is an op nobody can find.

## 11 The thermophysical property architecture

Property data in Choupo is *declarative systems with OpenFOAM-like explicit files*. A case does not hand the solver a bag of numbers; it *declares* its thermophysical system — `constant/thermoPhysPropDict (recordType thermophysicalPropertySystem; schemaVersion 2;)`, one record naming the components, the equilibrium formulation, the model in each phase slot, and the parameter files the run needs. The one-sentence contract:

*thermoPhysPropDict declares, ThermoPackageBuilder assembles, ThermoPackage computes; curation produces .dat, runtime does not estimate.*

### 11.1 The flow

|                      |                                                      |
|----------------------|------------------------------------------------------|
| thermoPhysPropDict   | the case DECLARES this (declarative)                 |
|                      |                                                      |
| ThermoPackageBuilder | runtime ASSEMBLY: loads + assembles, never estimates |
|                      | (src/thermo/ThermoPackageBuilder.cpp)                |
| ThermoPackage        | the mounted runtime COMPUTE object (unchanged)       |
|                      |                                                      |
| unit operations      | consume the ThermoPackage, as before                 |

*Terminology (load-bearing).* The runtime assembly step is the **ThermoPackageBuilder** — it **loads**. It is *never* a “resolver”: *resolver* is reserved for **curation-time** estimation (the Props Guide). Builder = runtime, loads; resolver = curation, may estimate; the two never mix.

### 11.2 The two axes

A substance never *is* molecular or electrolyte — the same NaCl plays different roles, and the role is chosen by the package a case selects, never stored on the substance. Two orthogonal choices define the role:

**Representation** (the *package* activates it): lumped → complete dissociation → partial / ion-pair → multi-ion. The complete, *K*-free split is the **dissociatesTo** ion stoichiometry on the component — a formula-like *identity*; the moment a split carries an equilibrium constant (an ion pair, a hydrate) it is *chemistry* and lives under **chemistry/**.

**Reference** (the *method* declares it): which datum the species anchor to — solid, pure-liquid Raoult, aqueous infinite-dilution, or fused-salt. **species/** records are medium-agnostic (one Na<sup>+</sup> file); the thermo data inside are medium-tagged blocks (**aqueousThermo**).

### 11.3 The data homes

`ls data/standards/` is the authority on this list, and that sentence is not a formality: this subsection was headed “the seven data homes” while the tree held eight directories, and the one it left out was **conventions/** — the versioned convention profiles the whole equilibrium-parameterisation identity references. A count written into prose is a derived number in a second home; it drifts, and it drifts silently. What follows is the *shape*, never the tally:

|                                        |                                                                                                                                                                                                                |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data/standards/                        |                                                                                                                                                                                                                |
| components/<name>.dat                  | identity (name, MW, nonvolatile)<br>+ dissociatesTo { Na 1; Cl 1; } [salts]<br>+ solidPhases{}: rho_p, k_v and the salt's<br>OWN dissolution anchor (the retired<br>phases/solid/<phase>.dat is a legacy read) |
| species/<name>.dat                     | one modelSpecies file per species:<br>formula + charge + medium-tagged thermo<br>(charge 0 for O2/N2(aq); never fed to a flowsheet)                                                                            |
| chemistry/<name>.dat                   | FLAT: REAL equilibria (stoichiometry + K + dH);<br>each record's recordType names its family<br>(aqueousSpeciation / gasLiquidEquilibrium /<br>ionExchangeEquilibrium) -- never a subfolder                    |
| parameters/                            |                                                                                                                                                                                                                |
| Pitzer/{pairs,lambda,psi}/...          | Pitzer binaries + ternaries                                                                                                                                                                                    |
| eNRTL/<c>-<a>.dat                      | eNRTL tau pairs                                                                                                                                                                                                |
| NRTL/ UNIQUAC/ Wilson/ Henry/ SRK/ ... | molecular pair catalogues                                                                                                                                                                                      |
| conventions/<profile>.dat              | the VERSIONED, immutable convention profiles<br>a curated parameterisation cites by name                                                                                                                       |

```

(Sander-Hxp-v1, PHREEQC-gasMolal-v1, ...):
what the constant MEANS, kept apart from
what it IS
assets/<name>.dat      FLAT with a 'kind' field -- construction
                        materials, membranes, adsorbents, resins:
mixtures/              one home, filtered by kind, not by directory
                        curated mixture records
utilities/<name>.dat   steam LP/MP/HP, cooling + chilled water,
                        dowthermA, hitecSalt, refrigerationPG,
                        electricity
(the property SYSTEM is not a directory at all: a case DECLARES it INLINE
 in constant/thermoPhysPropDict -- THE CENTRE. There is no shared package
 catalogue, and the old per-model methods/<name>.dat ceremony records were
 retired.)

```

`components/` stays *flat* (the loader resolves `<name>.dat` by exact name,  $O(1)$ ). There is deliberately **no** second component record per salt (no `apparent/` twin — one component, one file), no user-facing “true species” vocabulary (an ion is not *more real* than the salt; it is the speciation a model postulates — so: *model species*), and no `propertySets/` kind (it had zero readers and was removed rather than kept as ceremony). A minimal electrolyte system (active chemistry beside it in `constant/chemistryDict`):

```

recordType    thermophysicalPropertySystem;
schemaVersion 2;
components ( water NaCl );
equilibrium
{
    formulation electrolyteGammaPhi;
    aqueous
    {
        solvent          water;
        apparentComponents ( NaCl );
        activityModel { model Pitzer; }
        compositionBasis molality;
    }
    vapour { fugacityModel idealGas; }
}

```

The solid equilibrium activates through the chemistry selection (the  $K_{sp}$  anchor comes from the component’s `solidPhases`); the Pitzer pair is resolved by cation–anion name from the parameter catalogue.

## 11.4 The builder contract

`ThermoPackageBuilder::build` dispatches on the declared `equilibrium.formulation`. For an electrolyte formulation it (1) reads the component list; (2) identifies the ONE salt as the component whose standards record carries `dissociatesTo`, and takes its identity (MW, role) from `components/<salt>.dat`; (3) classifies the ions by charge sign from each ion’s `species/<name>.dat`; (4) loads the crystal ( $\rho_p, k_v$ ) and the saturation anchor from the salt component’s own `solidPhases{}` block, by the phase name the package’s `chemistry.salts` declares (the retired `phases/solid/<phase>.dat` stays readable for old external cases only); (5) assembles the Pitzer/eNRTL kernel through the same `SaltFromCatalogue` helpers the legacy path used — byte-identical by construction; (6) mounts the `ThermoPackage`. **It loads and assembles; it does not estimate** — a missing record fails clear:

```

FATAL: electrolyteGammaPhi selected the Pitzer activity model,
      but required parameter pair Na-Cl was not found.

```

For a molecular formulation (`gammaPhi` with NRTL) the same entry point reads the binary pair from `parameters/NRTL/` and mounts the classical NRTL package — one stack, electrolyte and molecular alike. The reference conventions are the `standardState` keys on the phase/group blocks (`pureLiquid`, `infiniteDilution`); the old per-model `methods/<name>.dat` ceremony records were retired once the dispatch became fully native.

## 11.5 Status

The legacy salt-level electrolyte front-end is fully retired: the `electrolyte{}` component block is gone from every salt `.dat` (ion identity now derives from `dissociatesTo`; dissolution data live in `chemistry/salts/`),

`ElectrolyteActivity::configure()` and the model `pitzer;/model eNRTL`; factory registrations are deleted, and the electrolyte tutorials declare `electrolyteGammaPhi` systems. A molecular case is the same record degenerate — a `gammaPhi` block, no dissociation, no chemistry; lightweight is not legacy. Two electrolyte front-ends remain deliberately separate: *multi-ion speciation* (`pitzerHmw/davies/edwardsPitzer` via a `propsDict speciate` op — see §11.6) and the flat membrane/DSPM-DE data path. The mixed-solvent, segment-based eNRTL of Chen & Song (2004) ships as `ENRTLMixedSolvent` with a primary-data validation tutorial (`tutorials/props/electrolyte/enrtl_mixed_nacl_ethanol_esteso`).

## 11.6 Adding an AQUEOUS activity model — and the four things that are not the registration line

`AqueousActivity` (the `speciate` path) has a *nullary* factory, unlike the `ActivityModel` bases above, so a model whose parameters depend on the run must assemble them inside `evaluate()`. Adding `edwardsPitzer` in August 2026 exercised every way that can go wrong, and the list is worth having before you start, because *four of the five defects were invisible to a compiling, self-checking, unit-tested model*.

1. **Register it, or it is not a capability.** The model sat in the tree for a day: it compiled, its closed-form self-checks against the source paper passed, and *no case could reach it*, because it was missing from `AqueousActivity::registerBuiltin()`. No unit-level test can see this. Write the gate so it runs the model *through the engine* on a real case.
2. **Add it to the declared-name table too.** A case selects a model by a declared word (`model EdwardsPitzer;`), which `src/propertyOps/CasePackage.H` maps to the factory key. Keep the accepted list and the refusal message reading *one* table: the if-chain that preceded it held those two facts in two places, where a model added to one and not the other refuses by naming itself unimplemented.
3. **An `ActivityResult` outlives the frame that built it.** Its closures are called by the solver long after `evaluate()` returns, so they must be *self-contained*. The Edwards kernel captured its names and molalities by value and its parameter set by reference; the asymmetry was harmless while every caller outlived its result, and segfaulted the moment an adapter built the model as a temporary. Capture by value.
4. **Say what *your* model’s scope is.** The speciation banner was a two-way switch on one model name with everything else falling through to Davies’ sentence — so a new model inherited a description of a model that was not running, and both halves of it were false (the quoted trust band belonged to Davies; the “ $a_w$  from  $\phi = 1$ ” claim was contradicted by the run’s own  $a_w$ ). A banner keyed on one branch and defaulting for the rest is a lie waiting for its second author.
5. **Teach `bin/choupo-import` about your parameter home, in the same commit.** This is the one that can corrupt data rather than merely mislead. A sealed case is forbidden the installation catalogue, so a record home the importer’s dependency closure does not know is a record home the sealed case *loses*. The first sealed Edwards case kept 9 of its 28 pair parameters — the ones that need no record at all — and still ran, still converged, and answered with 19 terms missing. Nothing refused.

**Key idea. Running is not agreeing.** The importer used to validate a seal by re-running the staged case with the catalogue hidden and checking the exit code, and that is exactly the check the defect above passes. `validate_staged_agrees` now compares the staged run against the case’s own `expected` golden and refuses with the likely cause named, installing nothing. A case with no golden yet keeps the exit-code check and is *told* its answer is unverified — an unverifiable claim announced is not the same as one quietly not made.

Where the parameters themselves live is a separate decision, and the doctrine is unchanged: *the tree never stores a derivative*. Edwards measured one family of interaction parameters and estimates the rest from it by four rules; only the measured tables are on disk, and the estimated pairs are derived at assembly time in `EdwardsCatalogue.cpp` and printed with the rule that produced each one. Keeping the estimation out of the kernel is what lets the kernel stay a pure function of its parameters, testable against the paper’s own closed forms with no filesystem in sight.

## 12 Adding components, materials, reactions

These are *data-driven* — no C++ changes needed. WHERE each number belongs is not a matter of taste: it is decided by the data-governance doctrine ([docs/ai/data-doctrine.md](#), summarised in §21). Read that before adding or moving a value.

### 12.1 Components

Choupo's component data cascades through tiers, lowest precedence to highest: `local < standard < case < sector < unit` (`local` is the private, gitignored `data/local/` tier; every use is announced `[local] UNVERIFIED`).

- `data/standards/components/<name>.dat` — the **curated central catalogue** shipped with the simulator. Carries only **arity-1 intrinsic** data: `identity`, `critical`, `gasIdeal`, `liquidPure`, `solid`, `transport`, the `standardThermochemistry` datum, and any model-named parameter block (`pcsaft{}`, `uniquac{}`, `cosmo{}`). Treat it as read-only — the engine *refuses* to write here; a change is a reviewed curation act. A **pair** number (NRTL  $\tau$ , Henry,  $k_{ij}$ , a heat of solution) is **never** copied into the `.dat`; it lives in a catalogue, referenced by name.
- `<case>/constant/components/<name>.dat` (and the same at `<sector>/` and `<unit>/`) — **per-scope overlays** for sample-specific refinements (a particular powder's sorption isotherm, a laboratory measurement).

**Key idea.** The overlay merges block-by-block, NOT field-by-field. An overlay carrying `solid { rho_p 1610; }` replaces the *entire* `solid{}` block — the standard's `k_v` is *lost*, not deep-merged. A reference-state block is the atomic unit of physical meaning; mixing a sample's `rho_p` with the catalogue's `k_v` would be a silent hidden hybrid. **The curator must copy the whole block they refine.** (This corrects the pre-doctrine “field-by-field” behaviour.)

The file format is identical in both tiers:

```
component <name>;
mw      78.11;           // g/mol
Tc      562.05;          // K
Pc      48.95;           // bar
omega   0.2103;
Tb      353.24;          // K
Hvap_Tb 30720.0;         // J/mol at Tb
Vliq    89.4e-6;         // m^3/mol (Wilson)

vaporPressure
{
    model    Antoine;
    A        4.01814;
    B        1203.835;
    C        -53.226;
}

idealGasHeatCapacity
{
    model     polynomial;
    coeffs    ( -31.368  0.4746  -3.107e-4  8.524e-8 );
    Tmin      200.0;
    Tmax      1500.0;
}

liquidHeatCapacity
{
    model     polynomial;
    coeffs    ( 162.94  -0.3493  9.51e-4  0.0 );
    Tmin      278.0;
    Tmax      500.0;
}

standardThermochemistry
{
```

```
dHf_298      82930.0;      // J/mol (NIST ideal-gas value)
s_298         269.20;      // J/(mol*K) (third-law absolute)
phase        gas;          // phase in which dHf_298 / s_298 are tabulated
}
```

Source the values from DIPPR / Reid-Prausnitz-Poling and *cite the source in a top-of-file comment* so they stay auditable.

**The phase field (mandatory in standardThermochemistry).** The reference state for formation enthalpy is documented in the Theory Guide (chapter on the elements datum). The **phase** keyword tells the solver in *which phase* **dHf\_298** is tabulated, so **Component::h\_formation(T, targetPhase)** can build the right integration path through Hvap / Hfus jumps and the matching  $c_p$  leg. Three values are allowed:



| phase  | required data                                                                                                               | typical species                                                                                                  |
|--------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| gas    | <code>idealGasHeatCapacity</code> , plus <code>HvapTb+Tc+Tb</code> if the liquid leg of any stream is needed                | <i>the convention — all 45 standard volatiles + radicals; follows NIST / JANAF</i>                               |
| liquid | <code>liquidHeatCapacity</code> ; <code>HvapTb+Tc+Tb</code> if any stream goes through the gas leg                          | reserved for compounds whose $H_f$ is published in the condensed datum (rare)                                    |
| solid  | <code>liquidHeatCapacity</code> (used as the dissolved-solute approximation $c_p^{\text{solid}} \approx c_p^{\text{liq}}$ ) | crystalline non-volatiles whose $H_f$ cannot be referenced to gas because the compound never vaporises (sucrose) |

The default is **gas** for backward compatibility with the earliest component files, but **new components should always declare it explicitly** so the datum is unambiguous at the source of truth. A **solid** or **liquid** declaration without the matching  $c_p$  model throws a clear error from `Component::h_formation` the first time a stream containing the component is evaluated.

**Common pitfall.** A missing `standardThermochemistry` is now fatal, not papered over. The energy ledger runs on the elements datum *only* (§22); the old “fall back to the sensible datum with a note” path was deleted. A component that appears in a stream the balance touches but carries no formation datum throws, naming the component — a loud curation gap, never a quietly wrong closure. Curate the eleven historical gaps (or the new component’s datum) before relying on the balance.

Add the new component name to the case’s `thermoPhysPropDict components (...)` list (or to a per-case `constant/components/<name>.dat` overlay if the data is sample-specific) and the new entry is picked up automatically — no recompile needed.

## 12.2 Materials

`data/materials/<name>.dat`:

```
material <name>;
rho      7900.0;      // kg/m^3
F_M      1.0;         // Guthrie material factor
sigma_y   138.0e6;    // Pa
maxT      700.0;      // K
maxP      50.0;       // bar
```

`MaterialRegistry::loadFrom()` slurps the directory on simulator start. No registration step needed.

## 12.3 Utilities

`data/standards/utilities/<name>.dat` catalogues a plant utility (steam header, cooling-water loop, hot-oil loop, refrigeration brine). The same registry pattern as materials and membranes: the `UtilityCatalogue::loadFrom(dataRoot)` call in each main slurps the directory at startup; the stream parser then resolves `utility <name>;` keys against it.

```
utility      steamLP;
tier         heating;           // heating | cooling
mechanism    condensation;      // condensation | sensible | evaporation

components   ( water );
state        saturatedVapour;
P            2.5 bar;
T_in         400.3 K;
T_out        400.3 K;           // condensate same T as steam

dutyPerKg    2186000.0;         // J/kg delivered (Hvap_water at T_sat)
cost         12.0;              // EUR/GJ delivered
costYear     2022;
description  "Low-pressure saturated steam header";
```

To add a new utility, drop a new `.dat` into the directory — no C++ change. When the utility loop carries a non-water fluid (oils, salts, brines), the `components` list points at a corresponding entry under `data/standards/components/`; the catalogue ships `dowthermA`, `hitecSalt` and `propyleneGlycol30` pseudo-components for that purpose. See the user guide for the case-author syntax `utility <name>;` and the `utilities` report that aggregates kg/h, MW, and EUR/h per category.

## 12.4 Reactions

In a case’s `constant/reactions`:

```
reactions
{
    ethanol_to_water_test
    {
        stoichiometry
        {
            ethanol    -1.0;
            water       +1.0;
        }
        kinetics
    }
}
```

```
    {  
      type      powerLaw;  
      A         1.0e6;      // s-1  
      E         80000.0;    // J/mol  
      orders    { ethanol 1.0; }  
    }  
  }  
}
```

Referenced from a CSTR / PFR by name (`reaction ethanol_to_water_test;`).

**Key idea.** The reactions library cascades UP, like the thermophysical system. When a branch of a fractal plant (§19) is run *standalone*, its named-reaction library resolves by walking up the parent-folder chain (capped at six levels) exactly as the thermo package does, and **announces its origin** — **reactions library: loaded -- LOCAL vs inherited from <parent>/....** Before this symmetry was restored, a sub-sector run alone died with “reaction not in constant/reactions” because only the local folder was consulted: the two cascades were asymmetric. No-silent-crutch: the cascade is loud.

## 13 Adding an outer driver / post-processor

### 13.1 Outer driver

src/outerDriver/MyDriver.{H,cpp} inherits OuterDriver. Override:

```
class MyDriver : public OuterDriver {
public:
    void readFromDict(const DictPtr&) override;
    int run() override;           // calls simulator_(flowsheetDict_) N times
};
```

The simulator functor and dict templates are injected by main.cpp:

```
auto driver = OuterDriver::New(outerDict);
driver->setSimulator      (simulate);           // SimulationResult(DictPtr)
driver->setFlowsheetDict  (flowsheetDict);      // mutable flowsheet template
driver->setThermoDict     (thermoDict);         // mutable thermoPhysPropDict
driver->run();
```

The simulator closure captures the thermo dict by `shared_ptr`, so any in-place `setScalarAtPath()` the driver makes on `thermoDict_` is picked up on the next `simulator_()` call.

Inside `run()`, you typically:

1. Mutate `flowsheetDict_>setScalarAtPath(...)` to change a sweep variable, OR `thermoDict_>setScalarAtPath(...)` to change a model parameter (e.g. NRTL `a_ij`).
2. Call `simulator_(flowsheetDict_)` → get a `SimulationResult`.
3. Extract KPIs / stream fields, accumulate residuals / responses.
4. Decide next step (LM / Nelder-Mead / GA / ...).
5. Eventually report and return 0 (converged) or 1 (failed).

Two concrete examples to read first:

- src/outerDriver/SweepDriver.{H,cpp} (~ 150 LOC) — 1-D sensitivity sweep.
- src/outerDriver/FitBinaryPair.{H,cpp} (~ 300 LOC) — Marquardt LM with FD Jacobian, mutates both flowsheet (composition) and thermo (pair params).

Register your driver in `OuterDriver::registerBuiltins()`.

### 13.2 Post-processor

src/postProcessing/MyPass.{H,cpp} inherits PostProcessor. Single method:

```
int run(SimulationResult&) override;
```

Append to `result.sizings` / `result.costs`, write CSVs, print to stdout. Register in `PostProcessor::registerBuiltins()`.

`PostProcessor::buildChain(postDict)` walks `postDict` top-level keys and instantiates one pass per recognised key. Add yours to the chain-builder in `PostProcessor.cpp`.

## 14 Adding a sizing model

`EquipmentSize` is a sub-factory used by `SizingPass`. Add `src/postProcessing/sizing/MyEquip.{H,cpp}` inheriting `EquipmentSize`:

```
class MyEquip : public EquipmentSize {
public:
    EquipmentSizing size(const std::string& unitName,
                        const SimulationResult& result,
                        const Material& material,
                        const DictPtr& designRules) override;
};
```

Use `result.streams.at(...)` and `result.kpis.at(unitName).at(...)` to get the process data, plus `designRules` for the design specs (pressure design,  $F_M$ , joint efficiency, ...). Fill `EquipmentSizing`:

```
EquipmentSizing d;
d.unitName      = unitName;
d.equipmentType = "myEquip";
d.material      = material.name;
d.values["V_R"] = ...;          // canonical size
d.values["t_wall"] = ...;       // m
d.values["weight"] = ...;       // kg
return d;
```

Register in `EquipmentSize::registerBuiltins()`.

The `Guthrie` costing model reads `d.equipmentType` to dispatch to the right correlation — so if your new equipment needs new cost coefficients, edit `src/postProcessing/costing/Guthrie.cpp` accordingly.

## 15 Coding conventions

### 15.1 Style

- C++ 17. `std::variant`, `std::optional`, `std::unique_ptr`, `std::filesystem` — yes. No raw `new` / `delete`. No `goto`.
- **Brace style:** Allman / open-brace on a new line for function definitions; Egyptian (`if (x) {`) for control flow.
- **Indent:** 4 spaces. No tabs anywhere.
- **Naming:** `snake_case` for locals, `camelCase` for methods, `PascalCase` for types, `UPPER_SNAKE` only for macros (we use exactly one: include guards via `#pragma once`).
- Compile cleanly with `-Wall -Wextra -Wpedantic` and fix anything that fires.

### 15.2 Includes

- Headers compile in isolation — every `.H` includes what it needs.
- Forward-declare aggressively in headers; pull full definitions only in `.cpp`.
- The build does not use precompiled headers.

### 15.3 Errors

- User-facing: throw `std::runtime_error` with a message a student can understand. Don't `assert` on user-data invariants.
- Programmer-facing (impossible state): `assert(...)` is fine.
- Never silently swallow errors. If a sub-step fails, surface it.

### 15.4 Comments

- Default to writing *no* comments. Identifiers should be obvious.
- Equations and references *deserve* a comment. Cite the source (Reid-Prausnitz-Poling page, DIPPR ID, Henley-Seader chapter).
- Top-of-file banner is preserved on every source — do not delete.

### 15.5 Header banner

Every `.H` and `.cpp` starts with:

```
/*-----*\
|  Choupo                                     |
|                                           |
|  Open-source chemical process simulator   |
|                                           |
|  Copyright (C) 2026 Vítor Geraldês         |
|  SPDX-License-Identifier: GPL-3.0-or-later |
|-----*/

Class
    Choupo::<ClassName>

Description
    <one-paragraph description>
/*-----*/
```

Don't omit this on new files — it doubles as licence notice.

## 16 Regression testing

Regression is run with `bin/runTests`. Run it before every change set; all previously-passing tutorials must still pass.

```
bin/runTests          # every tutorial (4 binaries + buildCode cases)
bin/runTests <case>   # one case
bin/runTests --record <case> # refresh that case's golden-master 'expected'
```

`runTests` does two checks, both stronger than a plain exit-code loop:

1. **NaN / inf guard — every case.** The structured result block is scanned for NaN or inf values; a hit FAILS the case. This is the check a bare `binary > /dev/null && PASS` loop *cannot* do: a solver can return NaN with exit code 0, and the old loop reported it green. That exact trap once hid a broken `membrane01` (NaN KPIs, exit 0) after a pressure-unit refactor.
2. **Reference KPIs — cases with an expected file.** Each line pins a KPI or stream value to its golden-master value with a relative tolerance; any deviation FAILS, printing `got X expected Y`.

**Key idea.** The `expected` file lives in the case directory. Format (`#` comments allowed):

| # kind | name    | key      | expected      | reitol |
|--------|---------|----------|---------------|--------|
| kpi    | flash01 | V_over_F | 0.30398357314 | 1e-6   |
| stream | vapor   | F        | 0.00844398814 | 1e-6   |

`kind` is `kpi` (inside the named unit's KPIs) or `stream` (inside the named stream). Pin the numbers worth guarding; `-record` writes them from the current run once you trust it. A case with no `expected` still gets the NaN guard.

At, four canonical cases ship an `expected` (`flash01`, `cstr01`, `column01`, `membrane01`); the rest run under the smoke + NaN guard. Add an `expected` whenever a case's numbers are load-bearing.

## 17 Debugging

### 17.1 Debug build

```
make MODE=debug all
gdb build/linux64GccDebug/choupoSolve
(gdb) r tutorials/flash01_benzene_toluene
```

The debug build leaves a separate symlink `choupoSolve.debug` so the release binary stays usable in parallel.

### 17.2 Verbosity

`controlDict { verbosity 4; }` enables debug-level prints — every Newton iteration, every  $K$ -value evaluation, every Wegstein step. For LL flashes that misconverge, watch for stagnation:  $g(x)$  should shrink monotonically near the root.

### 17.3 Sanitizers

Edit `make/compiler.mk` for the debug branch:

```
ifeq ($(MODE),debug)
    CXXFLAGS += -fsanitize=address,undefined -fno-omit-frame-pointer
    LDFLAGS  += -fsanitize=address,undefined
endif
```

Then rebuild. Useful when a unit-op produces NaN out of nowhere or when adding a new factory and a dispatch crashes.

### 17.4 The most common failure: NaN propagation

Symptoms: `verbosity 3` shows clean inputs, then  $V/F = \text{nan}$  mid-flash. Almost always a  $\log P_i^{\text{sat}}$  with  $P_i \leq 0$  (Antoine outside its valid range) or an empty composition ( $F = 0$ ). Add asserts where you suspect it.



## 18 Architecture deep-dive — the three layers

*What you'll learn.* The contract between the outer driver, the simulator core, and the post-processor. Read this twice before adding cross-layer logic.

```
+-----+
| Layer 1: OuterDriver (src/outerDriver/) |
| Sensitivity sweep, optimisation, parameter estimation, MC. |
| Owns: simulator_func, flowsheetDict_ (mutable). |
| Mutates flowsheetDict_ between calls via setScalarAtPath(). |
+-----+
| calls |
| v |
+-----+
| Layer 2: Simulator core |
| runSimulation(flowsheetDict,...) |
| -> ThermoPackage::readFromDict(thermoDict, db) |
| -> Flowsheet::solve(flowsheetDict, thermo, verbosity) |
| -> SimulationResult { streams, kpis, converged } |
| Pure function: same inputs -> same outputs. |
+-----+
| result |
| v |
+-----+
| Layer 3: PostProcessor (src/postProcessing/) |
| Sizing -> equipment dimensions (V_R, A, t_wall, weight,...) |
| Costing -> Guthrie/Turton with CEPCI, USD->EUR |
| (future) Reporting -> PDF / HTML / CSV bundles |
| Augments SimulationResult; never re-enters the simulator. |
+-----+
```

### The contract:

- Layer 1 *never touches the simulator's internal state* — it only mutates the dict it owns and calls the functor.
- Layer 2 *is pure* — `runSimulation(d) == runSimulation(d)` for the same dict. This is what lets Layer 1 do finite-difference Jacobians and Levenberg-Marquardt cleanly.
- Layer 3 *never re-runs the simulator* — it only reads the result. If you need to re-run (e.g. for utility costing), do it via an outer driver.

**Key idea.** Rule of thumb: if a quantity can be computed from a converged stream table plus KPIs, it belongs in a post-processor. If it requires re-solving the flowsheet, it belongs in an outer driver.

## 19 The fractal flowsheet — nesting and flattenNode

*What you'll learn.* How a plant of sectors of units collapses to ONE flat solver problem, why the hierarchy is namespace-only, and how a branch runs standalone.

A Choupo case is a **fractal**: a composite *sector* nests leaf units (or further sectors), to any depth. But the solver never sees the tree. `flattenNode` walks the declared children and collapses the hierarchy into **one flat solver problem** with dotted `plant.sector.unit` names — the hierarchy is *authoring and namespace* only, never a recursive solver-within-a-solver. One flat Newton-on-tears recycle solves the whole plant.

**Key idea.** The hierarchy is discovered from the *dicts*, not the filesystem. `flattenNode` iterates the `children(...)` declared in `flowsheetDict`; a child loads as a sector only when its own `flowsheetDict` exists. This is why the numeric instant directories (§20) sitting beside the sector folders are never mistaken for sectors — they are not declared children and have no `flowsheetDict`.

**Lean fractal folders + the dual-reader.** A node's topology may live either way, both first-class:

- the classic `<node>/system/flowsheetDict`, or
- the lean `<node>/flowsheetDict` directly at the node root.

A **dual-reader** keeps both loading: the child loader reads `<child>/flowsheetDict` first and *falls back* to `<child>/system/flowsheetDict`; the top-level `choupoSolve` does the same at the case root. Existing fractal cases (esterification, `twoSectorDemo`) load byte-identically through the fallback — **no forced mass-migration**.

**A branch runs alone.** Because both `thermoPhysPropDict` and the reactions library cascade UP the parent chain (§12), pointing `choupoSolve` at a sub-sector folder runs that branch as a complete case, inheriting the parent's thermo and reactions and announcing what it inherited.

**The model-boundary rule.** A per-unit `thermo {}` override REPLACES the global models for that unit (components stay global, via `thermoFor`). A stream crossing such a boundary is re-interpreted there. The governing law: ***H is the conserved truth, T is the model-dependent readout***. A model boundary is not a physical device, so there is no real  $\Delta T$  to absorb; H-continuity is a convention, not a law. The default is **hold-T, let-H-jump** (the discontinuity is visible in the printed enthalpy); a silent “hold *H*, flex *T*” is rejected. The honest opt-in is a **model-boundary audit** — print  $\Delta H = H_{\text{down}} - H_{\text{up}}$  at fixed  $(T, P, z)$ , fold it into the first-law ledger, and hard-refuse across any phase / `vf` / speciation flip. Full rationale: `docs/ai/energy.md`.

## 20 Solution instants — OpenFOAM-style per-iteration snapshots

*What you'll learn.* Where the converged (and intermediate) stream tables are written, and why the layout is faithful to OpenFOAM's time directories.

A run writes **durable per-iteration snapshots** of every stream, modelled on OpenFOAM's time directories. `src/io/SolutionWriter` owns the writes; the durability core is atomic (`.tmp` → `rename` + `fsync`, gated on `flushEach`), a restart re-seeds the recycle tear from the latest instant, and a litter sweep purges stale numeric dirs.

**The layout — instants at the CASE ROOT.** An instant is a bare iteration-*number* directory at the case root, beside `constant/`, `system/` and the fractal sector folders. “The time speaks for itself” — there is no `solution/` wrapper.

```
case/
  constant/ system/ reports/      # the usual case payload
  FERMENTATION/ CONCENTRATION/ ... # fractal SECTOR folders (alphabetic)
  0/                               # instant 0 (the initial seed)
    streams                        # ROOT flat stream table (the spine)
    FERMENTATION/ streams         # per-sector view
    CONCENTRATION/ streams
    byUnit/                       # per-unit breakout
  1/ 2/ 3/                        # later recycle iterations
  latest -> 3                     # convenience symlink to the highest
  solution.log                   # the instant index
```

**Key idea.** A stream IS a surface field ( $\phi$ ); the OpenFOAM analogy is exact. The payload file is named `streams`; `latest` is the highest-numbered instant (latestTime style — restart discovers it by the max integer dir name). A load-bearing *all-digit filter* on the numeric-dir scan guarantees a sector folder (alphabetic) is never swept or mistaken for an instant, and the fractal loader iterating declared children (not a filesystem walk) guarantees the reverse.

The pilot is `tutorials/plant/ChemicalPlantTutorial`; `writeInterval` controls how many instants land (set it to 1 on a 3-iteration case or you only see 0 and the final — and miss the whole tear-residual march).

## 21 The data cascade — arity, scope, kind

*What you'll learn.* The mechanical decision procedure for WHERE every thermodynamic number lives. This is the engine-side summary; the canonical, ratified reference is `docs/ai/data-doctrine.md` — read it before any curation act.

**Key idea.** A datum's home is decided by ARITY, then SCOPE, then KIND — in that order. Read the quantity's *definition*, not its frequency of use.

### (1) Arity — how many species does the definition name?

- **Arity-1** (the molecule alone on the table, no partner named — *not even implicitly*) → INTRINSIC, `data/standards/components/<name>.dat`,  $T_c, P_c, \omega$ , MW, `standardThermochemistry`, `solid`, `liquidPure`, `transport`.
- **Arity-2** (the definition must name a partner — a solvent, a counter-ion, an interaction species, *even water*) → PAIR/SET data in a **catalogue**, keyed by the pair, cited per value, referenced by name. NRTL  $\tau$ , Henry  $H$ , Pitzer  $\beta$ , EoS  $k_{ij}$ , a heat of solution. **Never copied into the .dat.**

**(2) Scope — at what level is the number TRUE?** A datum lives at the *highest level where it is true* (the canonical value is a curation act in `data/standards/`). A lower node overlays only when it makes the value *more* true — a sample-measured refinement. Precedence, low to high: `local` < `standard` < `case` < `sector` < `unit` (`local` = the private, announced `data/local/` tier).

**(3) Kind — the molecule, or the molecule-in-a-machine?** A *rate* (kinetics) or a *geometry / distribution* (PSD) is NOT component data at all — it lives in the operation's `constant/` (`constant/crystallisation`, `constant/dryingKinetics`, `constant/reactions`). PSD is a *stream* attribute.

**The aqueous TIER and the default-solvent resolver.** Water gets *no* exemption from the arity test: an “in-water” property is pair data. Its one earned privilege is a single canonical, by-name aqueous reference tier — `data/standards/species/<ion>.dat` for ions, and `data/standards/solution/<solute>-<solvent>.dat` for molecular solutes (the heat of solution). The package declares a default solvent once (`solvent water;`); when an in-solvent property is needed and no solvent is named at the call site, the resolver uses the default **and announces it every run**:

```
[thermo] dHsoln(sucrose): solvent not named -> DEFAULT = water;
          solution/sucrose-water.dat [Putnam & Kilday 1986]
```

Off-default it *fails with a remedy* (“provide `solution/sucrose-ethanol.dat` or set `solvent water;` explicitly”) — it never silently substitutes the water number. This is implemented in `ThermoPackage` (`solventName_`, the `SolutionRegistry` walk).

**The overlay merges block-by-block.** As in §12: a reference-state block (`solid{}`, `gasIdeal{}`, `liquidPure{}`, each `eosParameters.<model>{}`) is the atomic unit of physical meaning. An overlay replaces the whole block — the curator copies the whole block they refine; a lone-scalar overlay is the forbidden hidden hybrid.

**Model extensibility — `eosParameters{ <model>{...} }`.** The same definition test, asked of the *model*:

1. A model needing only the corresponding-states triad  $\{T_c, P_c, \omega\}$  (every cubic: SRK, PR, ...) **reads the triad off the Component and derives its own  $a_c, b, m, \alpha(T)$  in source** — adding it touches *zero* data files (this is why adding PR after SRK touched no `.dat`). Do not pre-bake  $a_c$  into the `.dat`.
2. A model-specific PURE parameter that cannot be generated (PC-SAFT's  $m, \sigma, \varepsilon/k$ ) goes in a `model`-keyed sub-block on the component, `eosParameters{ PCSAFT{...} }`, each factory reading *its own* sub-block and ignoring the rest.
3. PAIR parameters ( $k_{ij}$ ) go in a `model`-keyed catalogue, `data/standards/eos/<model>/binaryInteractions/<i>-<j>.dat`, with the always-permitted inline override in the package.

A model with no required `eosParameters.<model>` block **fails with a remedy**, never a silent corresponding-states fallback.

## 22 The elements datum — one enthalpy surface

**What you'll learn.** Why the energy balance runs on ONE datum, how the heat of reaction / solution / crystallisation emerge from it, and what throws when a datum is missing.

A heat balance that has to close *through* a reactor, a crystalliser, or a dissolution cannot use a sensible (arbitrary- $T_{\text{ref}}$ ) datum — it would silently drop the heat of reaction. Choupo runs the energy ledger on the **elements datum**: formation enthalpies at 298.15 K, the same reference as the ideal-gas  $H_{ig}$ , so the reaction / phase-change / dissolution enthalpy *emerges* as the difference of formation enthalpies.

**Key idea. One datum, no silent fallback (Vitor's law: elements-only, throw-on-gap).** The second datum and every silent path were deleted — `streamH_sensible`, `solidH_sensible`, the `tryDatum(elements = false)` fallback, the `EnergyRef::Sensible` arm. `streamH_elements` now *propagates a real missing-Hf gap*, naming the component, instead of swallowing it. A curation gap is loud, never a quietly wrong closure.

**The solid term.** A stream's enthalpy is computed over both phases: the fluid (`s.F`, `s.z`) **and** the solid carried in `s.s[]` (kmol/s per component), on the same elements datum —  $\sum_i s.s[i] \cdot h_i^{\circ}(\text{solid}, T)$ , the same solid formation leg `Component::h_formation(T, "solid")` uses. Before this, the crystals a crystalliser had just made were silently dropped from the stream enthalpy and the heat of crystallisation read zero; counting `s.s[]` (plus the aqueous solution tier for the dissolved solute) closed the crystalliser balance.

**Utility legs vs process streams — no double-count.** The ledger partitions each unit's ins/outs by category (`src/reporting/BalanceMath.H`, `EnergyBalanceReport.cpp`): utility-tagged legs (steam  $\rightarrow$  condensate) feed a `qBoundary` accumulator — the real heat crossing the boundary — while process streams feed  $h_{\text{in}}/h_{\text{out}}$ . An evaporator's `duty_kW` is delivered *through its own steam stream*, so it is INTERNAL and must NOT also be summed as an external item; a process-to-process exchanger's duty is likewise internal. Closure =  $100 \cdot \Delta H / \text{supplied}$  over genuine boundary heat only. This killed a fake 2131 % non-closure that was a pure double-count, not lost latent heat.

**One enthalpy surface everywhere.** The published stream `.H` (Flowsheet) and the isothermal-flash duty (`IsothermalFlash.cpp`) now run on the *same* canonical elements-datum form the report reads (`H_stream_formation`,  $h_{ig} - \Delta h_{\text{vap}}(T)$ ), not a Watson-at-298 +  $\int c_p^{\text{liq}}$  variant. The two differed by the liquid latent model and surfaced as a  $\sim 24\%$  gap between a flash's reported  $Q$  and its own stream  $\Delta H$ ; unifying the surface moves only the reported *duty* (an energy KPI), never the material flows.

## 23 Extending Choupo — step by step

*What you'll learn.* Two complete, copy-pasteable walkthroughs — *tim-tim-por-tim-tim* — each ending in a green `bin/runTests`. Recipe A: a new unit operation, end to end. Recipe B: a new property-prediction method, with the validation-first step that earns your trust before you simulate with it.

### 23.1 Recipe A — a new unit operation, end to end

**The worked example: `isothermalSplitter`** — a unit that splits one feed into two outlets by a fraction  $\theta$  to the top, both outlets at the feed's  $T, P, z$  (composition unchanged; the spec is the split fraction). Trivially simple physics on purpose — so every *mechanical* step is visible.

**Step 1 — The header.** `src/unitOperations/mixer/IsothermalSplitter.H` (it belongs beside the existing `Splitter`):

```
#pragma once
#include "core/Dictionary.H"
#include "thermo/ThermoPackage.H"
#include "unitOperations/UnitOperation.H"

namespace Choupo {

class IsothermalSplitter : public UnitOperation {
public:
    IsothermalSplitter() = default;
    const std::string& type() const override { return type_; }

    int solve(const DictPtr& dict,
              const ThermoPackage& thermo,
              int verbosity) override;

    std::vector<ProcessStream> producedStreams() const override { return produced_; }
    std::map<std::string, scalar> kpis() const override { return kpis_; }

private:
    inline static const std::string type_ = "isothermalSplitter";
    std::vector<ProcessStream> produced_;
    std::map<std::string, scalar> kpis_;
};

} // namespace Choupo
```

Note the exact `solve` signature — `int solve(const DictPtr&, const ThermoPackage&, int)` — and that `type()` returns a reference to a member string. Prepend the licence banner (§on header banner); it doubles as the licence notice.

**Step 2 — The implementation.** `src/unitOperations/mixer/IsothermalSplitter.cpp`. The Flowsheet injects the inlet and your parameters as sub-dicts of `dict`; you publish outlets into `produced_` in the order your `outputs (...)` declares:

```
#include "mixer/IsothermalSplitter.H"
#include "core/Dimensions.H"
#include <iostream>

namespace Choupo {

int IsothermalSplitter::solve(const DictPtr& dict,
                             const ThermoPackage& thermo,
                             int verbosity)
{
    auto feed = dict->subDict("feed");           // F (kmol/s SI), Tfeed, Pfeed
```

```

    auto comp = dict->subDict("composition"); // mole fractions z_i
    auto op    = dict->subDict("operation");   // YOUR parameters

    const scalar F      = feed->lookupScalar("F",      Dims::molarFlow);
    const scalar Tfeed   = feed->lookupScalar("Tfeed",  Dims::temperature);
    const scalar Pfeed   = feed->lookupScalar("Pfeed",  Dims::pressure);
    const scalar theta   = op->lookupScalar("splitFraction"); // [-], 0..1

    if (theta < 0.0 || theta > 1.0)
        throw std::runtime_error(
            "isothermalSplitter: splitFraction must be in [0,1]");

    std::vector<scalar> z(thermo.n());
    for (std::size_t i = 0; i < thermo.n(); ++i)
        z[i] = comp->lookupScalar(thermo.comp(i).name());

    if (verbosity >= 3)
        std::cout << " [isothermalSplitter] F = " << (F * 3600.0)
            << " kmol/h, split " << theta << " to top\n";

    produced_.clear(); // ORDER == outputs(...) order
    ProcessStream top; top.name = "top";
    top.F = theta * F; top.T = Tfeed; top.P = Pfeed; top.z = z;
    produced_.push_back(top);

    ProcessStream bot; bot.name = "bottom";
    bot.F = (1.0 - theta) * F; bot.T = Tfeed; bot.P = Pfeed; bot.z = z;
    produced_.push_back(bot);

    kpis_.clear(); // KPIs feed sizing/costing/sweep
    kpis_["F_top"]      = top.F;
    kpis_["F_bottom"]   = bot.F;
    kpis_["splitFraction"] = theta;

    return 0; // 0 == converged
}

} // namespace Choupo

```

**Key idea.** Fill EVERY field of a produced stream (F, T, P, z, and vf / s if relevant). A half-filled stream propagates NaN downstream and the NaN guard fails the case — by design.

**Step 3 — Register it (the one line).** Edit `src/unitOperations/UnitOperation.cpp`: add the include beside the other mixer includes, and one line in `registerBuiltin()`:

```

#include "mixer/IsothermalSplitter.H" // (1) beside #include "mixer/Splitter.H"
//...
void UnitOperation::registerBuiltin()
{
    auto reg = [](const std::string& name, auto&& maker) {
        registerType(name, std::forward<decltype(maker)>(maker)); };
    //...
    reg("splitter",      []{ return std::make_unique<Splitter>(); });
    reg("isothermalSplitter", []{ return std::make_unique<IsothermalSplitter>(); }); // (2)
    //...
}

```

That is the *entire* registration story — no macro, no static initialiser. A student greps `registerBuiltin` and sees your type listed.

**Step 4 — Build (the recursive pickup).**

```
make all
```

The Makefile discovers sources with `find src -name '*.cpp'`, so the new `.cpp` is compiled with no Makefile edit; `-MMD -MP` tracks the new header dependency automatically.

**Step 5 — Rebuild the WASM (MANDATORY).**

```
make wasm-gui          # = choupoSolve + choupoProps, the two the GUI uses
```

**Common pitfall.** Skip this and the GUI throws `UnitOperation::New: unknown type 'isothermalSplitter'` even though the native binary runs the type fine. The WASM solver is a separate Emscripten compile; a new built-in only reaches the browser after a WASM rebuild (`make wasm-gui`). The incremental build invalidates on source, flag and standards-content changes — a “Nothing to be done” after a real change is a graph bug, not something to paper over with ritual cleans (`make wasm-clean` stays available as deliberate diagnostics). Never run two `make wasm` concurrently — they clobber `gui/public/wasm/`.

**Step 6 — A minimal tutorial case.** Keep it trivial so it doubles as a regression test. `tutorials/steady/mixer/splitter01`

```
splitter01_isothermal/
splitter01.cho          # GUI marker (openable; may carry GUI layout)
system/
  controlDict           # application choupoSolve; verbosity 3;
  flowsheetDict         # the topology below
constant/
  thermoPhysPropDict    # v2 record: components + gammaPhi ideal
```

The `system/flowsheetDict` (one feed boundary, one unit, two product boundaries):

```
boundary
{
  inlets ( feed );
  feed { F 1.0 kmol/s; Tfeed 350 K; Pfeed 1 bar;
        composition { benzene 0.5; toluene 0.5; } }
}

units
(
  {
    name      S1;
    type      isothermalSplitter;
    in        feed;
    outputs   ( top bottom );      // ORDER matches produced_ order
    operation { splitFraction 0.3; }
  }
);
```

**Step 7 — Add KPIs to the regression (expected).** Your `kpis()` already feed sizing, costing and the sweep/optimisation drivers; pin the load-bearing ones to golden masters. Drop a `tutorials/steady/mixer/splitter01_isothermal/e`

| #      | kind | name     | key | expected | reltol |
|--------|------|----------|-----|----------|--------|
| kpi    | S1   | F_top    | 0.3 | 1e-6     |        |
| kpi    | S1   | F_bottom | 0.7 | 1e-6     |        |
| stream | top  | F        | 0.3 | 1e-6     |        |

then record/verify and run the full gate:



```
bin/runTests --record tutorials/steady/mixer/splitter01_isothermal # write from a trusted run
bin/runTests # the whole corpus must
    ↪ stay green
```

**Key idea.** The gate is `bin/runTests`, not a bare exit-code loop — NaN/inf guard on *every* case plus the expected KPI/stream comparison (§16). A solver can return NaN with exit code 0; only the structured-result scan catches it.

## 23.2 Recipe B — a new property-prediction method

A property model plugs into one of four bases. **Pick the base by the quantity you predict**, then follow the same `fromDict + registerModel` pattern — and validate against measured data *before* you trust it.

| You predict...                 | Base               | registerModel factory signature                   |
|--------------------------------|--------------------|---------------------------------------------------|
| $P^{\text{sat}}(T)$            | VaporPressureModel | (const DictPtr&)                                  |
| $c_p(T)$                       | HeatCapacityModel  | (const DictPtr&)                                  |
| $\gamma_i(T, x)$               | ActivityModel      | (const DictPtr&, const std::vector<std::string>&) |
| $\varphi_i, Z, H^{\text{res}}$ | EquationOfState    | (const DictPtr&, const std::vector<Component>&)   |

**B.1 — A vapour-pressure model (the simplest base).** Say you add a two-parameter Clapeyron  $P^{\text{sat}}$ .

1. `src/thermo/vaporPressure/Clapeyron.{H,cpp}`, inheriting `VaporPressureModel`, override the pure virtual scalar `Psat_Pa(scalar T_K) const`; a constructor `Clapeyron(const DictPtr&)` reads the parameters (mirror `Antoine.cpp`, whose ctor is `Antoine::Antoine(const DictPtr&)`).
2. Register — include + one line in `VaporPressureModel::registerBuiltins()`:

```
#include "Clapeyron.H"
//...
void VaporPressureModel::registerBuiltins()
{
    registerModel("Antoine",
        [](const DictPtr& d) -> std::unique_ptr<VaporPressureModel>
        { return std::make_unique<Antoine>(d); });
    registerModel("AmbroseWalton",
        [](const DictPtr& d) -> std::unique_ptr<VaporPressureModel>
        { return std::make_unique<AmbroseWalton>(d); });
    registerModel("Clapeyron",
        [](const DictPtr& d) -> std::unique_ptr<VaporPressureModel>
        { return std::make_unique<Clapeyron>(d); });
}
```

3. Select it in a component's `vaporPressure{ model Clapeyron; ... }` block.

**B.2 — Validate FIRST (the AAD step).** A property model is **not trustworthy until it has been checked against measured data**. Choupo's `choupoProps` binary is the validation weapon: run the *measured* dataset through the model and read the average absolute deviation (AAD) before you let the model into a flowsheet.

```
choupoProps <validationCase> # e.g. a propertyScan1D over measured T
# the run prints, per point: model vs measured, dev %; headline:
# AAD = 1.8 % AAD tolerance = 3.0 % -> PASS
```

**Key idea.** The number that earns trust is the AAD against measured data, not that the model compiles. The bench prints the per-point deviation and a headline AAD against a tolerance (the same machinery `HeatTransferBench` and the Pitzer activity validation use); a model that misses its anchor beyond tolerance fails loudly. This is the property half of “see, then decide” — you simulate with a model *because* you have seen its AAD, never because a dropdown recommended it.

**B.3 — An equation of state (and the data-doctrine path).** An EoS is the one base where WHERE the parameters live is a doctrine decision. The canonical reference is `docs/ai/data-doctrine.md §4`; in short:

1. **Corresponding-states EoS (reads only the triad).** If the model needs only  $\{T_c, P_c, \omega\}$ , **read them off the Component and derive the rest in source** — touch *zero* data files. This is exactly what SRK and PR do: the SRK constructor reads `c.Tc()`, `c.Pc()`, `c.omega()` and computes  $a_c = \Omega_a R^2 T_c^2 / P_c$ ,  $b = \Omega_b R T_c / P_c$ ,  $m = m(\omega)$ . Do *not* pre-bake  $a_c$  into the `.dat` (it would duplicate a value the model owns and let the two drift).
2. **Model-specific PURE parameters (cannot be generated).** PC-SAFT's  $\{m, \sigma, \varepsilon/k\}$  live in a `model`-keyed sub-block on the component — additive, never disturbing the triad:

```
// inside the component .dat
eosParameters
{
  PCSAFT { m 2.4653; sigma 3.6478e-10; epsilon_k 287.35;
           provenance { origin regressed; method "Gross & Sadowski 2001"; } }
  // SRK/PR appear in NO key here -- they read Tc, Pc, omega
}
```

Each factory reads *its own* `eosParameters.<model>` sub-block by name and ignores the rest, so one molecule carries SRK + PR + PC-SAFT simultaneously, each with its own `provenance{}`.

3. **Pair parameters  $k_{ij}$ .** A model-keyed catalogue, `data/standards/eos/<model>/binaryInteractions/<i>-<j>.dat`, with the always-permitted inline override. `SRK::fromDict` already reads an inline `binaryInteractions` list of `{ i; j; kij; }` dicts and resolves names positionally against the component list.

**The explicit factory, and the open follow-up.** Register the EoS the same explicit way, dispatching to a `fromDict` static (the `Component`-list overload):

```
registerModel("PCSAFT",
  [](const DictPtr& dict, const std::vector<Component>& comps)
    -> std::unique_ptr<EquationOfState>
  { return PCSAFT::fromDict(dict, comps); });
```

**Common pitfall. Glass-box requirement:** an EoS with no required `eosParameters.<model>` block on a component must **fail with a remedy** (“PCSAFT needs `m`, `sigma`, `epsilon_k` for `X`; fit them with `choupoProps` or pick SRK/PR which run off `Tc`, `Pc`, `omega`”), *never* a silent corresponding-states fallback. **Open follow-up** (noted in the doctrine): a uniform `modelParams` accessor on `Component` that hands a factory its `eosParameters.<model>` sub-dict by model name is not yet generalised — today each EoS that needs a pure block reads it itself. Until that lands, copy the `liquidViscosity{}` raw-sub-dict reader pattern.

## 24 Roadmap

Ordered by impact for the MCFT course and Vítor's research:

| Target    | Idea                                                                           | Difficulty     |
|-----------|--------------------------------------------------------------------------------|----------------|
| (shipped) | Levenberg-Marquardt parameter-estimation driver ( <code>fitBinaryPair</code> ) | —              |
|           | Nelder-Mead optimisation driver (minimise cost / TAC)                          | easy           |
|           | LL flash via 2nd-order Michelsen TPD with multiple restarts                    | hard           |
|           | Solids: inert phase + filter / cyclone / dryer unit-ops                        | medium         |
|           | Crystalliser + PSD basics                                                      | hard           |
|           | NRTL distillation robustness (Naphtali-Sandholm)                               | medium-hard    |
| v1.0      | Equation-oriented sparse Newton for the full flowsheet                         | very hard      |
| > v1      | Membrane NF/RO unit op (Vítor's research bridge)                               | research-grade |

### Tech-debt items also welcome:

- Reference logs for every tutorial → in-tree regression diff.
- Reaction-kinetics library (Arrhenius, Langmuir-Hinshelwood, Michaelis-Menten) as a `Kinetics` base class.

## 25 Things to NOT do

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*What you'll learn.* A short list, each item born from a real incident. Memorise it.

- Do **not** suggest a Python rewrite or wrapper. The course teaches numerical methods in C++; rewriting in Python would defeat the purpose.
- Do **not** add macro auto-registration. Explicit `registerBuiltin()` is non-negotiable (see §6).
- Do **not** add a heavy dependency — even a “small” one. Pedagogical visibility outweighs convenience.
- Do **not** split a binary by numerical strategy (*\*Foam*-style proliferation). Strategy is run-time. Splitting by *problem class* — the four binaries Solve / Batch / Ctrl / Props — is the deliberate exception, not a precedent for more.
- Do **not** bypass the dict format. Never use YAML, JSON, or TOML inside the case directory. The dict is the canonical wire format.
- Do **not** relitigate decisions in `CLAUDE.md` §10 without first reading what’s already been discussed and why.
- Do **not** merge code that breaks the regression, even if the failure is “obviously unrelated”.

## 26 Quick reference

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| Task                                    | Command / file                                 |
|-----------------------------------------|------------------------------------------------|
| Build release                           | <code>make all</code>                          |
| Build debug                             | <code>make MODE=debug all</code>               |
| Clean                                   | <code>make clean / make distclean</code>       |
| Source env                              | <code>source etc/bashrc</code>                 |
| Run a case                              | <code>runCase tutorials/&lt;name&gt;</code>    |
| Run from inside case dir                | <code>runCase</code>                           |
| List tutorials                          | <code>listCases</code>                         |
| List built-in unit-op types (in code)   | <code>UnitOperation::availableTypes()</code>   |
| Tweak a dict field programmatically     | <code>dict-&gt;setScalarAtPath(path, v)</code> |
| Add a unit op (full walkthrough)        | §23.1 (quick: §7)                              |
| Add a property model (full walkthrough) | §23.2 (quick: §9)                              |
| Add a component / material / reaction   | §12                                            |
| Add an outer driver / post-processor    | — (see the outer-driver section)               |
| Where does a number live?               | §21 + <code>docs/ai/data-doctrine.md</code>    |
| Energy datum / balance closure          | §22                                            |
| Fractal flowsheet + instants            | §19, §20                                       |
| Project conventions / AI rules          | <code>CLAUDE.md</code>                         |
| User-facing how-to                      | <code>docs/userGuide.pdf</code>                |

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*Colophon.* This manual was typeset in Latin Modern with the Choupo shared LaTeX preamble. Sources live in `docs/`; rebuild with `cd docs && make`.